

GRACE LUSENO

PROJECT 1

Artificial Immune System in Traffic Management

This project explores the application of Artificial Immune Systems (AIS) is a computational approach inspired by the by the human immune system which has the ability to learn, adapt, and identify threats. The study focuses on how adaptive and intelligent systems can be used to detect, respond to and manage dynamic traffic conditions which is a concept can be applied to traffic management in several ways.

Objectives

- i. Evaluate if AIS algorithms can learn and adapt to real-time traffic data to improve traffic light timing in a simulated environment.
- ii. To compare AIS-based traffic control with traditional methods by examining how well AIS systems reduce traffic congestion and travel times compared to fixed timing plans and basic sensor systems in a simulated traffic network.
- iii. To identify the most effective AIS algorithms for traffic optimization by determining which algorithms achieve the best performance in optimizing traffic flow in a simulated environment.
- iv. Analyse how different AIS parameter settings affect system performance in managing traffic flow in a simulated environment.

Ai BRANCHES

1. Machine Learning

- **Detect unusual traffic patterns** like congestion, accidents, or sudden flow changes, allowing for quicker response.
- **Optimize traffic light timing** by learning from data and adjusting dynamically to improve flow and reduce congestion.
- **Create decentralized control systems** where individual intersections manage traffic based on local conditions, improving overall efficiency.
- **Design self-healing networks** that automatically reroute traffic around disruptions, minimizing impact and maintaining flow.

Justification

- **Adaptive and dynamic solutions** that can respond to changing traffic conditions.
- **Robustness against unexpected disruptions** due to the decentralized nature of the system.

- **Improved efficiency and reduced congestion** through data-driven decision making.

It's important to note that AIS is a complex field and further research is needed to fully realize its potential in traffic management. However, it presents an exciting avenue for creating intelligent and adaptable transportation systems in the future where further research is needed to fully realize their potential.

TOOLS USED

Python

Database